Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/01/2026
PI Name:	Mason, Christopher Ph.D.		
Project Title:	The Impact of Spaceflight and Radiation on Clonal Hematopoiesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	chm2042@med.cornell.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	203-668-1448
Organization Name:	Weill Medical College of Cornell University		
PI Address 1:	1305 York Ave, Room Y13-04, Box 140		
PI Address 2:			
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10021-5663	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	05/01/2023	**End Date:**	04/30/2027
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 04/30/2027 per J. McFather/JSC (Ed., 5/11/26).		
Key Personnel Changes/Previous PI:	Per the FY26 report, Duane Hassane is no longer included in the list of Co-Investigators (Ed., 5/28/26).		
COI Name (Institution):	Bailey, Susan Ph.D. (Colorado State University) De Vlaminck, Iwijn Ph.D. (Cornell University) Elemento, Olivier Ph.D. (Weill Medical College of Cornell University) Levine, Ross MD (Sloan Kettering Institute for Cancer Research)		
Grant/Contract No.:	80NSSC23K0832		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Clonal hematopoiesis (CH) occurs when blood cells harboring an advantageous mutation propagate faster than others. These mutations can confer a risk for certain hematological cancers and cardiovascular disease. However, little is known about the methods for reducing these clones, nothing is known about CH in astronauts, and there is little information about the proper way to understand and stratify the risk of these mutations that are found in normal individuals. The goal of this proposal is to delineate the risk of CH for astronauts and other patients using improved, novel methods that map nucleic acids isolated from peripheral blood and urine. We hypothesize that radiation imparts a selective advantage, specifically in hematopoietic cells with DNA Damage Response mutations, and that spaceflight duration and radiation dose can drive and mediate this selection process. We will use a cohort of >1,100 patients from Weill Cornell Medicine (WCM) with both blood and urine, where 11% already show CH, as well as existing data from healthy cohorts (n=20,000) from the Englander Institute for Precision Medicine (EIPM) and Memorial Sloan Kettering Cancer Center (MSKCC) (n=25,000). We will use established methods for purifying and extracting DNA, using protocols developed for the NASA Twins Study and our published EIPM PreCISE-1 test for CH. This includes 100+ genes involved in clonal hematopoiesis, cancer, and cardiovascular diseases (CVD), sequenced to a depth of >2000x coverage (with unique molecular barcodes), giving >95% sensitivity for variant allele frequencies (VAF) > 1% (about 1/50 cells). The presence of CH can inform long-term CVD and cancer risk, and thus can serve as a useful metric for astronauts preparing for long-term missions. Also, CH confers a mutation-specific inflammatory risk, which can potentiate post-flight or post-radiation morbidity and be compared to other risk factors. Thus, identifying and managing CH represents a crucial unmet need for astronaut health, radiation risk stratification, and guiding oncology patients more broadly.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have updated the data portal with the latest CH data from the MSK-IMPACT cohorts, and published improved methods for variant detection from NGS data.
Task Progress:	Specific Aims The aims of this grant include two sets of aims, as a merger between a project originally funded for Twinstrand Biosciences (Dr. Jesse Salk, 2021) and Dr. Christopher Mason (2021). The merged set of aims are: Aim S1: Design and build a duplex sequencing (DS) assay to assess carcinogenesis via clonal hematopoiesis of indeterminate potential (CHIP) and bladder cancer driver genes Aim S2: Use DS to assess mutant clone size in CHIP targets and bladder cancer driver genes in astronaut samples prior to and after space exposure of different durations Aim S3: Determine if these carcinogenesis biomarkers are correlated to radiation exposure Mason Aim 2a: Provide context for CH mutations within two large cohorts (MSK and EIPM) Mason Aim 2b: Validate telomere length dynamics and cfDNA methylation in murine urine We completed the Salk aims for assay design (S1, S2) in 2023-25, working in concert with Dr. Salk and his team to migrate the technology to a new platform after the Twinstrand company was closed. Specifically, we used a custom capture probe set for CH and high-accuracy Element Biosciences chemistry for the DS assays, which was used for the publications in 2024-25 on the Inspiration4 crew and follow-up on the previously observed CH cohorts, specifically: 1. Garcia-Medina JS, Sienkiewicz K, Narayanan SA, Overbey EG, Grigorev K, Ryon KA, Burke M, Proszynski J, Tierney B, Schmidt CM, Mencia-Trinchant N, Klotz R, Ortiz V, Foox J, Chin C, Najjar D, Matei I, Chan I, Cruchaga C, Kleinman A, Kim J, Lucaci A, Loy C, Mzava O, De Vlaminck I, Singaraju A, Taylor LE, Schmidt JC, Schmidt MA, Blease K, Moreno J, Boddicker A, Zhao J, Lajoie B, Altomare A, Kruglyak S, Levy S, Yu M, Hassane DC, Bailey SM, Bolton K, Mateus J, Mason CE. Genome and clonal hematopoiesis stability contrasts with immune, cfDNA, mitochondrial, and telomere length changes during short duration spaceflight. Precision Clinical Medicine. 2024 Apr 8;7(1):pbae007. 2. Jones CW, Overbey EG, Lacombe J, Ecker AJ, Meydan C, Ryon K, Tierney B, Damle N, MacKay M, Afshin EE, Foox J, Park J, Nelson TM, Suhail Mohamad M, Byhaqui SGA, Aslam B, Tali UA, Nisa L, Menon PV, Patel CO, Khan SA, Ebert DJ, Everson A, Schubert MC, Ali NN, Sarma MS, Kim J, Houerbi N, Grigorev K, Garcia Medina JS, Summers AJ, Gu J, Altin JA, Fattahi A, Hirzallah MI, Wu JH, Stahn AC, Beheshti A, Klotz R, Ortiz V, Yu M, Patras L, Matei I, Lyden D, Melnick A, Banerjee N, Mullane S, Kleinman AS, Loesche M, Menon AS, Donoviel DB, Urquieta E, Mateus J, Sargsyan AE, Shelhamer M, Zenhausern F, Bershad EM, Basner M, Mason CE. Molecular and physiological changes in the SpaceX Inspiration4 civilian crew. Nature. 2024 Aug;632(8027):1155-64. 3. Corti G, Kim J, Enguita FJ, Guarnieri JW, Grossman LI, Costes SV, Fuentealba M, Scott RT, Magrini A, Sanders LM, Singh K, Sen CK, Juran CM, Paul AM, Furman D, Calleja-Agius J, Mason CE, Galeano D, Bottini M, Beheshti A. To boldly go where no microRNAs have gone before: spaceflight impact on risk for small-for-gestational-age infants. Communications Biology. 2024 Oct 5;7(1):1268. doi: 10.1038/s42003-024-06944-6. In the past year, we also improved the base-calling methods for whole-genome sequencing (WGS) and variant calling, by working closely with the National Institute of Standards and Technology (NIST), and their Genome in a Bottle (GAIB) Working group for genotyping improvements for NGS methods. This work led to one published paper and several others in preparation. The first paper (Daniels et al., 2025), developed a benchmark set of subclonal (mosaic) variants in the widely used HG002 Genome in a Bottle reference, addressing a major gap in genomics, which is the lack of reliable ground truth for low-frequency variants. We identified and validated mosaic single-nucleotide variants (SNVs) using deep sequencing (~300×) and multiple orthogonal technologies (Illumina, PacBio HiFi, BGI, Element), producing a curated and annotated set of variants with allele fractions typically in the ~5–30% range, but with a demonstration of ways to get below 1% or even 0.1% in the future. We also defined “clean” genomic regions with no detectable mosaic variation, enabling robust assessment of false positives. 1. Daniels CA, Abdulkadir AA, Cleveland MH, McDaniel JH, Jáspez D, Rubio-Rodríguez LA, et al. Characterization of subclonal variants in HG002 Genome in a Bottle reference material as a resource for benchmarking variant callers. Cell Genomics. 2025;6:101104. https://

Other related papers that are in-press or review in relation to these aims includes: 2. Nelson TM, Rose JK, Walter CE, Cervantes-Navarro GL, et al. Pharmacogenomics guided spaceflight: The intersection between space-flown drugs and space genes. bioRxiv. 2024. doi:10.1101/2024.01.16.575951.

3. Winer DA, Du H, Kim J, Chang V, Burke M, Winer S, Costes SV, Frippiat JP, Sams C, Paul AM, Wu H, Ullrich O, Baatout S, Beheshti A, Mason CE, Choukér A, Crucian BE. Astroimmunology: The effects of spaceflight and its associated stressors on the immune system. Nat Rev Immunol. 2026 Mar;26(3):189-212. doi: 10.1038/s41577-025-01226-6. Epub 2025 Oct 16. PMID: 41102553; PMCID: PMC12903758.

Aim S3 links to the other Aims (2a, 2b), wherein the latest CH variant data from MSKCC has been loaded onto the cBio portal for comparison across various cancers. Specifically, members of our team have integrated the CH data from a recent MKSCC paper (Stonestrom et al., 2024) into the cBio data portal, which can now be browsed online. One example is a means by which to browse data from cohorts in 2021, 2022, and 2023 (“Clonal hematopoiesis mutations identified in blood samples from patients whose tumor-blood pairs were analyzed using MSK-IMPACT”) with more recent patents with CH: https://

In total, there are now 67,517 patients on the combined CH portal, which can be browsed in terms of CH type, mutational allele frequency, clinical metadata, and patient sub-sets. A key feature of the dataset is the mutation landscape, where clonal hematopoiesis–associated genes dominate, including DNMT3A (14.6%), TET2 (4.5%), PPM1D (3.6%), and ASXL1 (1.8%), alongside canonical tumor suppressors such as TP53. The mutation count distribution is heavily skewed toward lower mutation burdens, with a smaller subset of highly mutated samples. Demographically, the cohort is skewed toward older individuals (peak around ages 60–75), with incomplete metadata for sex and race but a majority of samples lacking full annotation.

We are now in the process of ingesting and cleaning up the EIPM CH cohort at WCM, and will complete that work in 2026. We will also complete testing of the cfDNA annotations from urine in 2026.

Bibliography Type:	Description: (Last Updated: 07/13/2026)
Articles in Other Journals or Periodicals	Nelson TM, Rose JK, Walter CE, Cervantes-Navarro GL, Schmidt CM, Lin R, Alexander E, Zheng JT, Glicksberg BS, Schmidt JC, Overbey E, Rana B, Patel H, Schmidt MA, Mason CE. "Pharmacogenomics guided spaceflight: The intersection between space-flown drugs and space genes." bioRxiv preprint server. Posted January 26, 2024. https://doi.org/10.1101/2024.01.16.575951 , Jan-2024
Articles in Peer-reviewed Journals	Daniels CA, Abdulkadir AA, Cleveland MH, McDaniel JH, Jáspez D, Rubio-Rodríguez LA, Muñoz-Barrera A, Lorenzo-Salazar JM, Flores C, Yoo B, Ebrahim Sahraeian SM, Wang Y, Rossi M, Visvanath A, Murray L, Chen W-T, Catreux S, Han J, Mehio R, Parnaby G, Carroll A, Chang P-C, Shafin K, Cook D, Kolesnikov A, Brambrink L, Eeman Mootor MF, Patel Y, Yamaguchi TN, Boutros PC, Sienkiewicz K, Foox J, Mason CE, Lajoie BR, Ruiz-Perez CA, Kruglyak S, Zook JM, Olson ND. "Characterization of subclonal variants in HG002 Genome in a Bottle reference material as a resource for benchmarking variant callers." Cell Genom. 2026 Apr 8;6(4):101104. https://doi.org/10.1016/j.xgen.2025.101104 ; PubMed PMID: 41421359; PubMed Central PMCID: PMC13069850 , Apr-2026
Articles in Peer-reviewed Journals	Winer DA, Du H, Kim J, Chang V, Burke M, Winer S, Costes SV, Frippiat JP, Sams C, Paul AM, Wu H, Ullrich O, Baatout S, Beheshti A, Mason CE, Choukér A, Crucian BE. "Astroimmunology: The effects of spaceflight and its associated stressors on the immune system." Nat Rev Immunol. 2026 Mar;26(3):189-212. https://doi.org/10.1038/s41577-025-01226-6 ; PubMed PMID: 41102553; PubMed Central PMCID: PMC12903758 , Mar-2026
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Articles in Peer-reviewed Journals	Ferraro S, Sengupta S, Shaw J, Dave A, Cheli M, Colon R, Zuccotti G, Mason CE, Marcovina S. "Point of care testing of biochemical markers for monitoring astronauts during long duration missions in deep space." 2025 Oct 28;64(3):570-581. https://doi.org/10.1515/cclm-2025-0526 ; PubMed PMID: 41140086 , Oct-2025
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Articles in Peer-reviewed Journals	Willett JDS, Kim J, Sakharkar A, Park J, Agyemang AA, Saygili EA, Proszynski J, Nelson TM, Mason CE. "The data science of health in space." Annu Rev Biomed Data Sci. 2026 May 18. Review. Online ahead of print. https://doi.org/10.1146/annurev-biodatasci-092724-045605 ; PMID: 42149965 , May-2026
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Significant Media Coverage	Doglin E. "Into the deep: As humans return to the Moon, researchers are trying to understand—and thwart—the biological toll of deep-space radiation." Science Magazine, January 29, 2026. https://www.science.org/content/article/humans-return-moon-scientists-confront-dangers-deep-space-radiation , Jan-2026